

PENGWIN Task 1 v3.6 — A1

Progressive Affinity-ABBC

Algorithm Description · 2026-08-16

1. Submission status

v3.6-a1-progressive-affinity is a separate upload candidate. It keeps the v3.5 Stage-A model, hybrid random-forest family router, largest-component ROI policy, anatomy-specific Stage-B experts and all model weights unchanged. Only the deterministic instance decoder is replaced by the validation-selected A1 pipeline described below. The previous image/model pair remains available for rollback until the candidate passes the Grand Challenge container test.

2. Method overview

The submission performs per-fragment instance segmentation of pelvic and femoral fractures with a serial two-stage STU-Net-B cascade:

1. **Stage A — anatomy.** A 5-class whole-volume model identifies sacrum, left/right hipbone and femur.
2. **Hybrid family routing.** A random forest selects pelvic or femur when confident; the organizers' acquisition rule is the uncertainty tiebreak.
3. **Stage B — fracture experts.** The routed anatomy selects a Sacrum, shared-Left/Right-Hip or Femur expert. It emits four ABBC channels and nine learned same-instance affinity channels at 1-, 3- and 9-voxel offsets.
4. **A1 decoding.** Short-range affinity creates watershed supervoxels. Robust 3/9-voxel evidence vetoes weak RAG merges. Full ABBC then splits partitions missing a fracture boundary and merges interfaces unsupported by Boundary. Finally, candidate fragments from 1 to 5 cm³ require agreement from all three affinity ranges; larger candidates require two of three.
5. **Assembly.** Instances use the official ranges: sacrum 1-50, left hip 51-100, right hip 101-150 and femur 151-200.

3. Frozen model contract

Component	Deployed value
Stage A	Dataset539_PelvicFemurAnatomyV3, PenguinTrainerSTUNetBaseAnatomyV301, fold 0
Router	v3.3 hybrid RF, confidence margin 0.15

Component	Deployed value
Stage B dataset	Dataset538_PelvicFemurBICMFragmentV5
Stage B experts	...V308SacrumExpertDeployedVal, ...V308HipExpertDeployedVal, ...V308FemurExpertDeployedVal
Stage B fold/checkpoint	fold 0 / checkpoint_best.pth
Stage B output	13 channels: 4 ABBC + 9 affinity
Initial partition	1/3/9-voxel RAG-veto, T=0.75, minimum 32 range pairs
ABBC refinement	3 split passes, 400-voxel minimum piece, Boundary merge margin 0.05/3
A1 small-candidate rule	predicted smaller side 1,000-5,000 mm ³ : affinity 3/3; otherwise 2/3
A1 affinity evidence	mean same-instance affinity ≥ 0.55 and at least 8 observations per range

The v3.6 model archive is byte-identical to v3.5. This isolates the decoder change and avoids introducing a new training or weight-selection variable.

4. Model archive layout

Grand Challenge extracts the archive directly under `/opt/ml/model`:

```

/opt/ml/model/

├─ nnunet/results/

|   └─ Dataset539_PelvicFemurAnatomyV3/

|       └─ PenguinTrainerSTUNetBaseAnatomyV301___.../fold_0/

|           └─ Dataset538_PelvicFemurBICMFragmentV5/

```

```

|   └─ PenguinTrainerSTUNetBaseAffinityV308SacrumExpertDeployedVal___.../fold_0/

|   └─ PenguinTrainerSTUNetBaseAffinityV308HipExpertDeployedVal___.../fold_0/

|   └─ PenguinTrainerSTUNetBaseAffinityV308FemurExpertDeployedVal___.../fold_0/

└─ stage1_router/stage1_target_router_fold0.joblib

```

5. Local evaluation

All rows use the same frozen 68-patient / 132-anatomy official-aligned local proxy. They are not hidden-test leaderboard scores.

Decoder	Dice	HD 95 mm	AS SD mm	Recall	Precision
v3.5 short affinity	0.8 556 49	6.6 869 71	1.8 522 73	0.9 314 12	0.9 32 19 7
Aggressive Boundary-only	0.8 828 73	3.9 933 62	0.9 832 34	0.9 48 14 2	0.8 567 73
A0, affinity 2/3	0.8 831 47	3.6 51 58 3	0.9 01 11 4	0.9 437 23	0.8 852 90
v3.6 A1, small 3/3	0.8 85 31 6	3.6 555 91	0.9 014 34	0.9 462 48	0.8 854 17

Small-fragment (1-5 cm³) recall is 33/53 for A0 and 34/53 for A1; large fragment recall remains 262/267. A1 improves A0 Dice by 0.002169 and F1 by 0.001209 with three additional split errors. It does not reach the predeclared 35/53 small-fragment target, so v3.6 remains an independent hidden-score probe.

6. Reproducibility and limitations

- Stage A, router, ROI policy, Stage-B experts and model archive are frozen.
- The four decoder modules vendored in the container are the same sources used for the reported ablation; synthetic regression and deployment-contract tests cover the stage order and A1 thresholds.
- A1 was selected on the reported validation set. Hidden-test generalization and the 10-minute platform runtime must be checked before activation.
- The stronger Core/Border veto and mutual-best variants were rejected because they lowered local instance F1 and increased split errors.

7. Runtime and licensing

The image targets an NVIDIA T4 16 GiB GPU. Stage A and each Stage-B expert are loaded serially. STU-Net-B and nnU-Net are used under their respective open-source licenses; Stage-B initialization derives from the official TotalSegmentator base_ep4k checkpoint.

8. References

- Isensee, F. et al. nnU-Net, *Nature Methods* 18, 203-211 (2021).
- STU-Net / TotalSegmentator bone pretraining.
- PENGWIN 2026 Task 1 baseline:
github.com/YzzLiu/PENGWIN2026_Task1_AutoSeg_Baseline.